

Before start activity verify the AWS-CLI installed and configure

```
md_rustam@DESKTOP-CPK0PUB:~$ aws configure list
```

Name	Value	Type	Location
profile	<not set>	None	None
access_key	*****RX5I	shared-credentials-file	
secret_key	*****qzvZ	shared-credentials-file	
region	us-east-1	config-file	~/.aws/config

Step-01. Every AWS region has a already created default VPC. Verify it:

aws ec2 describe-vpcs --filters Name=isDefault,Values=true

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 describe-vpcs --filters Name=isDefault,Values=true
{
  "Vpcs": [
    {
      "OwnerId": "637423571410",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-065b3db9f63c673ac",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-0e55dc323224aeac9",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-0afb1770b1f5a640"
    }
  ]
}
```

Step 2: Each Availability Zone having subnets with the Default VPCs.

aws ec2 describe-subnets --filters Name=vpc-id,Values=vpc-0abc12345def67890

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 describe-subnets --filters Name=vpc-id,Values=vpc-0e55dc323224aeac9
{
  "Subnets": [
    {
      "AvailabilityZoneId": "us-east-1a",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "637423571410",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-034a896defab167e0",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-034a896defab167e0",
      "State": "available",
      "VpcId": "vpc-0e55dc323224aeac9",
      "CidrBlock": "172.31.16.0/20",
      "AvailableIpAddressCount": 4091,
      "AvailabilityZone": "us-east-1a",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    },
    {
      "AvailabilityZoneId": "us-east-1b",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "637423571410",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-034a896defab167e0",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-034a896defab167e0",
      "State": "available",
      "VpcId": "vpc-0e55dc323224aeac9",
      "CidrBlock": "172.31.16.0/20",
      "AvailableIpAddressCount": 4091,
      "AvailabilityZone": "us-east-1b",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    }
  ]
}
```

```

    "AvailabilityZoneId": "us-east-1a",
    "MapCustomerOwnedIpOnLaunch": false,
    "OwnerId": "637423571410",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-0eb6361ea4319d3a7",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "BlockPublicAccessStates": {
      "InternetGatewayBlockMode": "off"
    },
    "SubnetId": "subnet-0eb6361ea4319d3a7",
    "State": "available",
    "VpcId": "vpc-0e55dc323224aeac9",
    "CidrBlock": "172.31.32.0/20",
    "AvailableIpAddressCount": 4091,
    "AvailabilityZone": "us-east-1c",
    "DefaultForAz": true,
    "MapPublicIpOnLaunch": true
  },
  {
    "AvailabilityZoneId": "us-east-1a",
    "MapCustomerOwnedIpOnLaunch": false,
    "OwnerId": "637423571410",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-0252352dc0f972362",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "BlockPublicAccessStates": {
      "InternetGatewayBlockMode": "off"
    },
    "SubnetId": "subnet-0252352dc0f972362",
    "State": "available",
    "VpcId": "vpc-0e55dc323224aeac9",
    "CidrBlock": "172.31.80.0/20",
    "AvailableIpAddressCount": 4091,
    "AvailabilityZone": "us-east-1a",
    "DefaultForAz": true,
    "MapPublicIpOnLaunch": true
  },
  {
    "AvailabilityZoneId": "us-east-1a",
    "MapCustomerOwnedIpOnLaunch": false,
    "OwnerId": "637423571410",
    "AssignIpv6AddressOnCreation": false,
    "Ipv6CidrBlockAssociationSet": [],
    "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-0c768179ff8f67ca1",
    "EnableDns64": false,
    "Ipv6Native": false,
    "PrivateDnsNameOptionsOnLaunch": {
      "HostnameType": "ip-name",
      "EnableResourceNameDnsARecord": false,
      "EnableResourceNameDnsAAAARecord": false
    },
    "BlockPublicAccessStates": {
      "InternetGatewayBlockMode": "off"
    },
    "SubnetId": "subnet-0c768179ff8f67ca1",
    "State": "available",
    "VpcId": "vpc-0e55dc323224aeac9",
    "CidrBlock": "172.31.80.0/20",
    "AvailableIpAddressCount": 4091,
    "AvailabilityZone": "us-east-1a",
    "DefaultForAz": true,
    "MapPublicIpOnLaunch": true
  }
]

```

```

        "BlockPublicAccessStates": {
            "InternetGatewayBlockMode": "off"
        },
        "SubnetId": "subnet-0c768179ff8f67ca1",
        "State": "available",
        "VpcId": "vpc-0e55dc323224aeac9",
        "CidrBlock": "172.31.64.0/20",
        "AvailableIpAddressCount": 4091,
        "AvailabilityZone": "us-east-1f",
        "DefaultForAz": true,
        "MapPublicIpOnLaunch": true
    },
    {
        "AvailabilityZoneId": "use1-az3",
        "MapCustomerOwnedIpOnLaunch": false,
        "OwnerId": "637423571410",
        "AssignIpv6AddressOnCreation": false,
        "Ipv6CidrBlockAssociationSet": [],
        "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-0563a272c98fc3a87",
        "EnableDns64": false,
        "Ipv6Native": false,
        "PrivateDnsNameOptionsOnLaunch": {
            "HostnameType": "ip-name",
            "EnableResourceNameDnsARecord": false,
            "EnableResourceNameDnsAAAARecord": false
        },
        "BlockPublicAccessStates": {
            "InternetGatewayBlockMode": "off"
        },
        "SubnetId": "subnet-0563a272c98fc3a87",
        "State": "available",
        "VpcId": "vpc-0e55dc323224aeac9",
        "CidrBlock": "172.31.48.0/20",
        "AvailableIpAddressCount": 4091,
        "AvailabilityZone": "us-east-1e",
        "DefaultForAz": true,
        "MapPublicIpOnLaunch": true
    },
    {
        "AvailabilityZoneId": "use1-az1",
        "MapCustomerOwnedIpOnLaunch": false,
        "OwnerId": "637423571410",
        "AssignIpv6AddressOnCreation": false,
        "Ipv6CidrBlockAssociationSet": [],
        "SubnetArn": "arn:aws:ec2:us-east-1:637423571410:subnet/subnet-018aa6475fe6f509d",
        "EnableDns64": false,
        "Ipv6Native": false,
        "PrivateDnsNameOptionsOnLaunch": {
            "HostnameType": "ip-name",
            "EnableResourceNameDnsARecord": false,
            "EnableResourceNameDnsAAAARecord": false
        },
        "BlockPublicAccessStates": {
            "InternetGatewayBlockMode": "off"
        },
        "SubnetId": "subnet-018aa6475fe6f509d",
        "State": "available",
        "VpcId": "vpc-0e55dc323224aeac9",
        "CidrBlock": "172.31.0.0/20",
        "AvailableIpAddressCount": 4091,
        "AvailabilityZone": "us-east-1d",
        "DefaultForAz": true,
        "MapPublicIpOnLaunch": true
    }
}
]
}

```

Step 3: Create a Security Group

aws ec2 create-security-group --group-name MyDefaultSG --description "Allow SSH and HTTP" -- vpc-0e55dc323224aeac9

```

md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 create-security-group --group-name MyDefaultSG --description "Allow SSH and HTTP" --vpc-id vpc-0e55dc323224aeac9
{
  "GroupId": "sg-0a105760c3863d04b",
  "SecurityGroupArn": "arn:aws:ec2:us-east-1:637423571410:security-group/sg-0a105760c3863d04b"
}

```

aws ec2 authorize-security-group-ingress --group-id sg-0a105760c3863d04b --protocol tcp --port 22 -
-cidr 0.0.0.0/0

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 authorize-security-group-ingress --group-id sg-0a105760c3863d04b --protocol tcp --port 22 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-040908e246b98e4ef",
      "GroupId": "sg-0a105760c3863d04b",
      "GroupOwnerId": "637423571410",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:637423571410:security-group-rule/sgr-040908e246b98e4ef"
    }
  ]
}
```

aws ec2 authorize-security-group-ingress --group-id sg-0a105760c3863d04b --protocol tcp --port 80 -
-cidr 0.0.0.0/0

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 authorize-security-group-ingress --group-id sg-0a105760c3863d04b --protocol tcp --port 80 --cidr 0.0.0.0/0
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-01fa91e09a817b243",
      "GroupId": "sg-0a105760c3863d04b",
      "GroupOwnerId": "637423571410",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "0.0.0.0/0",
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:637423571410:security-group-rule/sgr-01fa91e09a817b243"
    }
  ]
}
```

Step 4: Create a new key pair with a new name to access the instance operating system.

aws ec2 create-key-pair --key-name MyKey2 --query "KeyMaterial" --output text > MyKey2.pem

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 create-key-pair --key-name MyKey2 --query "KeyMaterial" --output text > MyKey2.pem; ls -l MyKey2
-rw-r--r-- 1 md_rustam md_rustam 1679 May  3 11:29 MyKey2.pem
```

Step 5: Create 2 instance (Linux and Windows)

aws ec2 run-instances --image-id "ami-0ed094fb1304fd857" --count 1 --instance-type t2.micro --key-name MyKey2 --security-group-ids "sg-0a105760c3863d04b" --subnet-id "subnet-0252352dc0f972362" --associate-public-ip-address --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=LinuxInstance}]"

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 run-instances --image-id "ami-0ed094fb1304fd857" --count 1 --instance-type t2.micro --key-name MyKey2 --security-group-ids "sg-0a105760c3863d04b" --subnet-id "subnet-0252352dc0f972362" --associate-public-ip-address --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=LinuxInstance}]"
{
  "ReservationId": "r-06b0b31f1ffeab820",
  "OwnerId": "637423571410",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "c95c272e-cbfa-4eba-933e-a9ea2f3be639",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2026-05-03T11:31:12.000Z",
            "AttachmentId": "eni-attach-0fbf2534dbfe9e679",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0a105760c3863d04b",
              "GroupName": "MyDefaultSG"
            }
          ],
          "Ipv6Addresses": [],

```

```

        "MacAddress": "12:90:40:35:c1:21",
        "NetworkInterfaceId": "eni-06c12112aed71cc59",
        "OwnerId": "637423571410",
        "PrivateDnsName": "ip-172-31-94-103.ec2.internal",
        "PrivateIpAddress": "172.31.94.103",
        "PrivateIpAddresses": [
            {
                "Primary": true,
                "PrivateDnsName": "ip-172-31-94-103.ec2.internal",
                "PrivateIpAddress": "172.31.94.103"
            }
        ],
        "SourceDestCheck": true,
        "Status": "in-use",
        "SubnetId": "subnet-0252352dc0f972362",
        "VpcId": "vpc-0e55dc323224aeac9",
        "InterfaceType": "interface",
        "Operator": {
            "Managed": false
        }
    }
},
"RootDeviceName": "/dev/xvda",
"RootDeviceType": "ebs",
"SecurityGroups": [
    {
        "GroupId": "sg-0a105760c3863d04b",
        "GroupName": "MyDefaultSG"
    }
],
"SourceDestCheck": true,
"StateReason": {
    "Code": "pending",

```

```

    "Message": "pending"
},
"Tags": [
    {
        "Key": "Name",
        "Value": "LinuxInstance"
    }
],
"VirtualizationType": "hvm",
"CpuOptions": {
    "CoreCount": 1,
    "ThreadsPerCore": 1
},
"CapacityReservationSpecification": {
    "CapacityReservationPreference": "open"
},
"MetadataOptions": {
    "State": "pending",
    "HttpTokens": "required",
    "HttpPutResponseHopLimit": 2,
    "HttpEndpoint": "enabled",
    "HttpProtocolIpv6": "disabled",
    "InstanceMetadataTags": "disabled"
},
"EnclaveOptions": {
    "Enabled": false
},
"BootMode": "uefi-preferred",
"PrivateDnsNameOptions": {
    "HostnameType": "ip-name",
    "EnableResourceNameDnsARecord": false,
    "EnableResourceNameDnsAAAARecord": false
},

```

```

"CurrentInstanceBootMode": "legacy-bios",
"Operator": {
    "Managed": false
},
"InstanceId": "i-0a31896d8f6f77646",
"ImageId": "ami-0ed094fb1304fd857",
"State": {
    "Code": 0,
    "Name": "pending"
},
"PrivateDnsName": "ip-172-31-94-103.ec2.internal",
"PublicDnsName": "",
"StateTransitionReason": "",
"KeyName": "MyKey2",
"AmiLaunchIndex": 0,
"ProductCodes": [],
"InstanceType": "t2.micro",
"LaunchTime": "2026-05-03T11:31:12.000Z",
"Placement": {
    "AvailabilityZoneId": "us-east-1a",
    "GroupName": "",
    "Tenancy": "default",
    "AvailabilityZone": "us-east-1a"
},
"Monitoring": {
    "State": "disabled"
},
"SubnetId": "subnet-0252352dc0f972362",
"VpcId": "vpc-0e55dc323224aeac9",
"PrivateIpAddress": "172.31.94.103"
}
]
}

```

```
aws ec2 run-instances --image-id "ami-0798c9ebaf695e14e" --count 1 --instance-type t3.micro --key-name MyKey2 --security-group-ids "sg-0a105760c3863d04b" --subnet-id "subnet-0252352dc0f972362" --associate-public-ip-address --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=LinuxInstance}]"
```

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 run-instances --image-id "ami-0798c9ebaf695e14e" --count 1 --instance-type t3.micro --key-name MyKey2 --security-group-ids "sg-0a105760c3863d04b" --subnet-id "subnet-0252352dc0f972362" --associate-public-ip-address --tag-specifications "ResourceType=instance,Tags=[{Key=Name,Value=LinuxInstance}]"
```

```
{
  "ReservationId": "r-00613dbc6f068f85b",
  "OwnerId": "637423571410",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "1ce1bdd4-a822-4dcf-87fa-443e98b4c6bf",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2026-05-03T11:34:31.000Z",
            "AttachmentId": "eni-attach-0651027baebfad88a",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0a105760c3863d04b",
              "GroupName": "MyDefaultSG"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "12:72:18:ec:d3:7d",
          "NetworkInterfaceId": "eni-03f1051a820923949",
          "OwnerId": "637423571410",
          "PrivateDnsName": "ip-172-31-86-168.ec2.internal",
          "PrivateIpAddress": "172.31.86.168",
          "PrivateIpAddresses": [
            {
              "Primary": true,
              "PrivateDnsName": "ip-172-31-86-168.ec2.internal",
              "PrivateIpAddress": "172.31.86.168"
            }
          ],
          "SourceDestCheck": true,
          "Status": "in-use",
          "SubnetId": "subnet-0252352dc0f972362",
          "VpcId": "vpc-0e55dc323224aeac9",
          "InterfaceType": "interface",
          "Operator": {
            "Managed": false
          }
        }
      ],
      "RootDeviceName": "/dev/sda1",
      "RootDeviceType": "ebs",
      "SecurityGroups": [
        {
          "GroupId": "sg-0a105760c3863d04b",
          "GroupName": "MyDefaultSG"
        }
      ],
      "SourceDestCheck": true,
      "StateReason": {
        "Code": "pending",
```

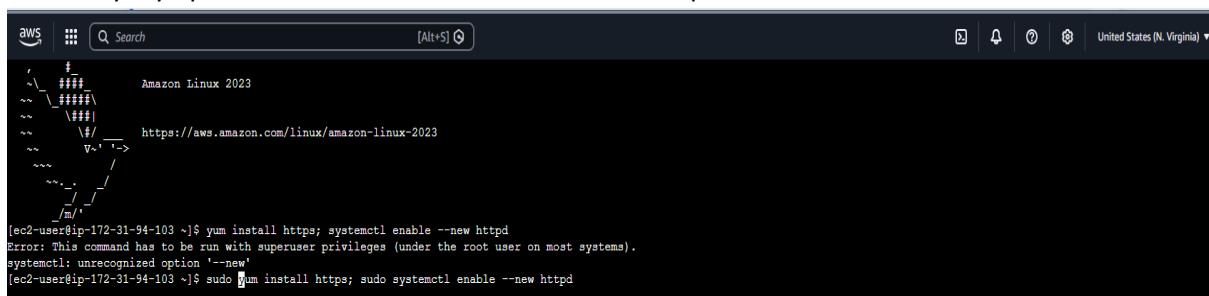
```

    "Message": "pending"
  },
  "Tags": [
    {
      "Key": "Name",
      "Value": "LinuxInstance"
    }
  ],
  "VirtualizationType": "hvm",
  "CpuOptions": {
    "CoreCount": 1,
    "ThreadsPerCore": 2
  },
  "CapacityReservationSpecification": {
    "CapacityReservationPreference": "open"
  },
  "MetadataOptions": {
    "State": "pending",
    "HttpTokens": "required",
    "HttpPutResponseHopLimit": 2,
    "HttpEndpoint": "enabled",
    "HttpProtocolIpv6": "disabled",
    "InstanceMetadataTags": "disabled"
  },
  "EnclaveOptions": {
    "Enabled": false
  },
  "BootMode": "uefi",
  "PrivateDnsNameOptions": {
    "HostnameType": "ip-name",
    "EnableResourceNameDnsARecord": false,
    "EnableResourceNameDnsAAAARecord": false
  },
  "Operator": {
    "Managed": false
  },
  "InstanceId": "i-0d95cb4aec4e24b3",
  "ImageId": "ami-0798c9ebaf695e14e",
  "State": {
    "Code": 0,
    "Name": "pending"
  },
  "PrivateDnsName": "ip-172-31-86-168.ec2.internal",
  "PublicDnsName": "",
  "StateTransitionReason": "",
  "KeyName": "MyKey2",
  "AmiLaunchIndex": 0,
  "ProductCodes": [],
  "InstanceType": "t3.micro",
  "LaunchTime": "2026-05-03T11:34:31.000Z",
  "Placement": {
    "AvailabilityZoneId": "use1-az2",
    "GroupName": "",
    "Tenancy": "default",
    "AvailabilityZone": "us-east-1a"
  },
  "Platform": "windows",
  "Monitoring": {
    "State": "disabled"
  },
  "SubnetId": "subnet-0252352dc0f972362",
  "VpcId": "vpc-0e55dc323224aeac9",
  "PrivateIpAddress": "172.31.86.168"
}
]
}

```

Step 5: Install Web Service on both the instances Linux (Amazon Linux 2) and Windows (Server 2025) access the instance with associated key pair of the both systems.

`ssh -i "MyKey2.pem" ec2-user@ec2-18-212-176-174.compute-1.amazonaws.com`



```

AWS Search [Alt+S]
United States (N. Virginia)

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-94-103 ~]$ yum install httpd; systemctl enable --new httpd
Error: This command has to be run with superuser privileges (under the root user on most systems).
systemctl: unrecognized option '--new'
[ec2-user@ip-172-31-94-103 ~]$ sudo yum install httpd; sudo systemctl enable --new httpd

```

sudo yum update -y; sudo yum install -y httpd
sudo systemctl enable --now httpd

```
[ec2-user@ip-172-31-94-103 ~]$ sudo yum install httpd; sudo systemctl enable --now httpd
Last metadata expiration check: 0:00:17 ago on Sun May  3 11:43:06 2026.
Dependencies resolved.

Package                               Architecture      Version            Repository         Size
----
Installing:
httpd                                x86_64            2.4.66-1.amzn2023.0.1  amazonlinux        47 k
Installing dependencies:
apr                                  x86_64            1.7.5-1.amzn2023.0.4  amazonlinux        129 k
apr-util                             x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        97 k
apr-util-ldap                        x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        13 k
generic-logos-httpd                 noarch            10.0.0-12.amzn2023.0.3  amazonlinux        19 k
httpd-core                           x86_64            2.4.66-1.amzn2023.0.1  amazonlinux        1.4 M
httpd-filesystem                     noarch            2.4.66-1.amzn2023.0.1  amazonlinux        13 k
httpd-tools                           x86_64            2.4.66-1.amzn2023.0.1  amazonlinux        81 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2  amazonlinux        315 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3  amazonlinux        33 k
Installing weak dependencies:
apr-util-openssl                     x86_64            1.6.3-1.amzn2023.0.2  amazonlinux        15 k
mod_http2                             x86_64            2.0.27-1.amzn2023.0.3  amazonlinux        166 k
mod_lua                               x86_64            2.4.66-1.amzn2023.0.1  amazonlinux        60 k

Transaction Summary
--
Install 13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Is this ok [y/N]: y

[ec2-user@ip-172-31-94-103 ~]$ systemctl status httpd
0 httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:httpd.service(8)

[ec2-user@ip-172-31-94-103 ~]$ sudo systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-94-103 ~]$ systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-05-03 11:48:09 UTC; 2s ago
     Docs: man:httpd.service(8)
  Main PID: 26971 (httpd)
    Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 1120)
  Memory: 13.1M
    CPU: 66ms
  CGroup: /system.slice/httpd.service
          └─26971 /usr/sbin/httpd -DFOREGROUND
            └─26972 /usr/sbin/httpd -DFOREGROUND
              └─26973 /usr/sbin/httpd -DFOREGROUND
                └─26974 /usr/sbin/httpd -DFOREGROUND
                  └─26975 /usr/sbin/httpd -DFOREGROUND

May 03 11:48:09 ip-172-31-94-103.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 03 11:48:09 ip-172-31-94-103.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
May 03 11:48:09 ip-172-31-94-103.ec2.internal httpd[26971]: Server configured, listening on: port 80
[ec2-user@ip-172-31-94-103 ~]$
```

Step 6: Create EBS Volume (5 GB) on both instances.

aws ec2 create-volume --availability-zone us-east-1a --size 5 --volume-type gp2 --tag-specifications "ResourceType=volume,Tags=[{Key=Name,Value=MyDataVolume}]"

```
wd_rustam@DESKTOP-CPK0PUB:~$ aws ec2 create-volume --availability-zone us-east-1a --size 5 --volume-type gp2 --tag-specifications "ResourceType=volume,Tags=[{Key=Name,Value=MyDataVolume}]"
{
  "AvailabilityZoneId": "use1-az2",
  "Iops": 100,
  "Tags": [
    {
      "Key": "Name",
      "Value": "MyDataVolume"
    }
  ],
  "VolumeType": "gp2",
  "MultiAttachEnabled": false,
  "VolumeId": "vol-0161351216b485921",
  "Size": 5,
  "SnapshotId": "",
  "AvailabilityZone": "us-east-1a",
  "State": "creating",
  "CreateTime": "2026-05-03T11:55:03.000Z",
  "Encrypted": false
}
```

Step 7: Attach EBS Volume to the both EC2 instances

Linux EC2

aws ec2 attach-volume --volume-id vol-0161351216b485921 --instance-id i-0a31896d8f6f77646 --device /dev/sdf

```
md_rustam@DESKTOP-CPK0PUB:~$ aws ec2 attach-volume --volume-id vol-0161351216b485921 --instance-id i-0a31896d8f6f77646 --device /dev/sdf
```

Windows EC2

```
aws ec2 attach-volume --volume-id vol-0161351216b485921 --instance-id i-0a31896d8f6f77646 --device xvdf
```

Step 8: Take Snapshot of the Volume

```
aws ec2 create-snapshot --volume-id vol-0161351216b485921 --description "Snapshot of MyDataVolume"
```

Step 9: Create New EBS Volume from Snapshot

```
aws ec2 create-volume --availability-zone ap-south-1a --snapshot-id snap-xxxxxxx --volume-type gp2 --tag-specifications "ResourceType=volume,Tags=[{Key=Name,Value=RestoredVolume}]"
```

Post completed the assigned tasks, committed successfully and pushed the changes to the GitHub repository.